

SQL DATABASE SYSTEMS REPORT

Data Types • ER Diagrams • OLTP & OLAP

01 SQL DATA TYPES

02 ER DIAGRAMS

03 OLTP & OLAP

Database Systems Coursework

Prepared for Academic Submission • 2026

01

SQL Data Types

Numeric, String, Date/Time, Boolean & More

02

ER Diagrams

Entities, Attributes, Relationships & Cardinality

03

OLTP vs OLAP

Definitions, Differences & Use Cases

SQL (Structured Query Language) data types define the kind of value that a column can store. Choosing the correct data type is essential for data integrity, storage efficiency, and query performance. SQL data types are broadly grouped into categories as described below.

1.1 Numeric Data Types

Numeric types store numbers — either exact integers or approximate floating-point values.

Data Type	Storage	Range / Notes	Example Use
INT / INTEGER	4 bytes	-2,147,483,648 to 2,147,483,647	StudentID, Age
TINYINT	1 byte	0 to 255 (unsigned)	Rating (1–5)
SMALLINT	2 bytes	-32,768 to 32,767	Quantity
BIGINT	8 bytes	Very large whole numbers	Transaction IDs
DECIMAL(p,s)	Variable	Exact fixed-point; p=precision, s=scale	Price, Salary
FLOAT / REAL	4 bytes	Approximate decimal (7 digits)	Scientific data
DOUBLE	8 bytes	Approximate decimal (15 digits)	GPS coordinates

Tip: Use DECIMAL for financial data — FLOAT can introduce rounding errors.

1.2 String / Text Data Types

String types store alphanumeric characters and text values.

Data Type	Max Size	Fixed/Variable	Example Use
CHAR(n)	255 chars	Fixed-length	Gender ('M'/'F'), Country code
VARCHAR(n)	65,535 chars	Variable-length	Name, Email, Address
TEXT	65,535 bytes	Variable (large)	Descriptions, Comments
MEDIUMTEXT	16 MB	Variable	Blog posts, Articles
LONGTEXT	4 GB	Variable	Books, Large documents
ENUM(...)	Up to 65,535 members	Fixed set of values	Status: 'Active','Inactive'

SET(...)	Up to 64 members	Multiple values allowed	Tags, Categories
----------	------------------	-------------------------	------------------

Remember: CHAR pads with spaces to fill fixed length. Use VARCHAR for most text fields to save storage.

1.3 Date and Time Data Types

Data Type	Format	Range	Example Use
DATE	YYYY-MM-DD	1000-01-01 to 9999-12-31	DateOfBirth, EnrollDate
TIME	HH:MM:SS	-838:59:59 to 838:59:59	ClassStartTime
DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 to 9999-12-31	OrderDateTime
TIMESTAMP	YYYY-MM-DD HH:MM:SS	1970-01-01 to 2038-01-19	LastUpdated (auto)
YEAR	YYYY	1901 to 2155	ManufactureYear

1.4 Boolean & Binary Data Types

BOOLEAN / BOOL

Stores TRUE (1) or FALSE (0). Used for flags like IsActive, IsDeleted. In MySQL, stored as TINYINT(1).

BINARY(n)

Fixed-length binary data. Stores raw bytes. Used for storing hashed passwords.

BLOB

Binary Large Object. Stores images, audio, or any binary file. Variants: TINYBLOB, MEDIUMBLOB, LONGBLOB.

BIT(n)

Stores n-bit integer values. Useful for compact binary flags. BIT(1) = single boolean bit.

1.5 Other Important Types

Data Type	Description	Example Use
JSON	Stores JSON formatted text natively (MySQL 5.7+)	Config data, API responses
GEOMETRY	Stores spatial/geographic data	Maps, Location data
UUID / CHAR(36)	Universally unique identifier string	Distributed systems IDs
SERIAL	Auto-increment alias for BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	Primary keys

Best Practice: Always choose the SMALLEST data type that safely fits your data. This saves disk space, speeds up queries, and improves index performance in MySQL.

An Entity-Relationship (ER) Diagram is a visual blueprint of a database. It shows the entities (tables), their attributes (columns), and the relationships between them. ER diagrams are the first step in designing a well-structured, normalized database.

2.1 Core Components

Symbol / Component	Shape	Description	Example
Entity	Rectangle	Represents a real-world object or concept stored as a table	STUDENT, COURSE
Attribute	Oval / Column	Property or characteristic of an entity	StudentName, DOB
Primary Key Attribute	Underlined Oval	Uniquely identifies each record in the entity	StudentID
Relationship	Diamond	Describes how entities are connected	Enrolls, Teaches
Weak Entity	Double Rectangle	Cannot exist without a related strong entity	ENROLLMENT detail
Multi-valued Attribute	Double Oval	Can hold multiple values	PhoneNumbers
Derived Attribute	Dashed Oval	Calculated from other attributes	Age from DOB

2.2 Cardinality (Relationship Ratios)

Cardinality defines how many instances of one entity relate to instances of another.

1 : 1 One-to-One

One entity instance relates to exactly one other. Example: Person ↔ Passport

1 : N One-to-Many

One instance relates to many others. Example: Department → Employees

M : N Many-to-Many

Many instances relate to many others. Example: Students ↔ Courses (via Enrollment table)

2.3 Sample ER Diagram — Student Enrollment System

The diagram below models a university system. A STUDENT enrolls in many COURSEs (M:N), captured via an ENROLLMENT entity. Both STUDENT and COURSE belong to a DEPARTMENT.

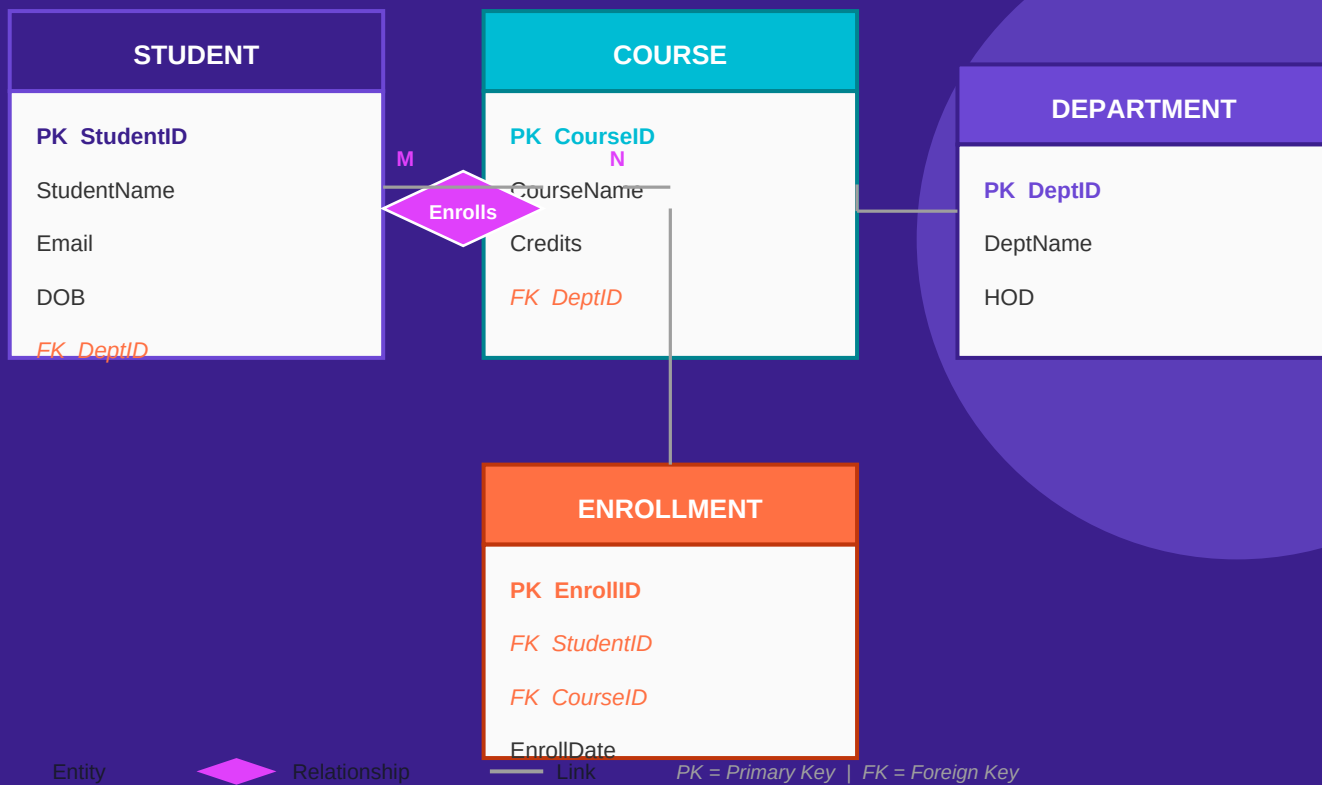


Fig. 1 — ER Diagram: University Student Enrollment System

2.4 From ER Diagram to SQL Tables

Once the ER diagram is finalized, each entity becomes a table in the database. Relationships become foreign keys or junction tables.

STUDENT Table

```
CREATE TABLE STUDENT ( StudentID INT PRIMARY KEY AUTO_INCREMENT, StudentName VARCHAR(100) NOT NULL, Email VARCHAR(150) UNIQUE, DOB DATE, DeptID INT, FOREIGN KEY (DeptID) REFERENCES DEPARTMENT(DeptID) );
```

ENROLLMENT Table

```
CREATE TABLE ENROLLMENT ( EnrollID INT PRIMARY KEY AUTO_INCREMENT, StudentID INT NOT NULL, CourseID INT NOT NULL, EnrollDate DATE DEFAULT CURRENT_DATE, Grade CHAR(2), FOREIGN KEY (StudentID) REFERENCES STUDENT(StudentID), FOREIGN KEY (CourseID) REFERENCES COURSE(CourseID) );
```

Key Rule: Every M:N relationship in an ER diagram must be resolved into a junction/bridge table (like ENROLLMENT) with foreign keys referencing both parent entities.

Two fundamental database processing models exist in modern data systems: **OLTP (Online Transaction Processing)** and **OLAP (Online Analytical Processing)**. Understanding the difference is critical for designing the right database system for the right purpose.

OLTP Online Transaction Processing

Manages day-to-day operational transactions. Handles large numbers of short, fast queries. Designed for INSERT, UPDATE, DELETE operations. Example: Banking, E-commerce, Hospital records.

OLAP Online Analytical Processing

Supports complex analysis and decision-making. Handles large volumes of historical data. Designed for SELECT with aggregations and JOINS. Example: Business Intelligence, Data Warehousing.

3.1 Detailed Comparison

Feature	OLTP	OLAP
Full Form	Online Transaction Processing	Online Analytical Processing
Purpose	Manage daily business transactions	Analyze historical data for decisions
Operations	INSERT, UPDATE, DELETE, SELECT	Primarily complex SELECT queries
Query Type	Simple, short, fast queries	Complex, long-running analytical queries
Data Volume	Current / recent data (GBs)	Historical data (TBs to PBs)
Database Design	Highly normalized (3NF)	Denormalized (Star/Snowflake schema)
Response Time	Milliseconds	Seconds to minutes
Users	Clerks, cashiers, operational staff	Data analysts, managers, executives
Concurrency	Thousands of concurrent users	Few users, but heavy queries
Backup	Regular, frequent backups critical	Less frequent; data rarely changes
Examples	ATM, POS systems, booking apps	Data warehouse, BI dashboards

Analogy: OLTP is like a cashier processing one customer at a time (fast & transactional). OLAP is like a manager reviewing all sales data from the past 5 years (broad & analytical).

3.2 OLTP — Key Characteristics

OLTP systems are optimized for **high throughput of small transactions**. They use ACID properties to ensure data reliability.

Atomicity

Each transaction is all-or-nothing. If any step fails, the whole transaction rolls back.

Consistency

Transactions bring the DB from one valid state to another, respecting all rules.

Isolation

Concurrent transactions execute independently, preventing dirty reads.

Durability

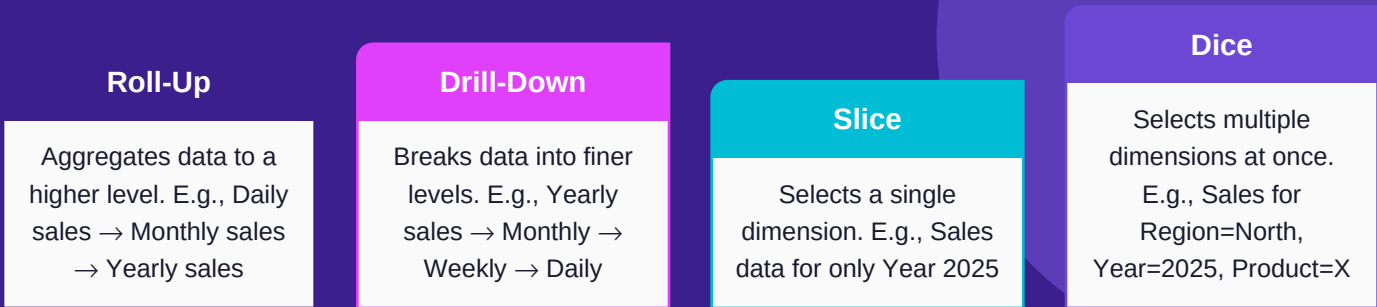
Once committed, a transaction persists even in case of system failure.

Typical OLTP Query (Bank Transfer):

```
START TRANSACTION; UPDATE ACCOUNT SET Balance = Balance - 5000 WHERE AccID = 101; UPDATE  
ACCOUNT SET Balance = Balance + 5000 WHERE AccID = 202; INSERT INTO TRANSACTION_LOG (FromAcc,  
ToAcc, Amount, TxnDate) VALUES (101, 202, 5000, NOW()); COMMIT;
```

3.3 OLAP — Key Characteristics

OLAP systems use **multidimensional data models** — often stored in a data warehouse. They support operations like drill-down, roll-up, slice, and dice.



Typical OLAP Query (Business Intelligence):

```
SELECT Region, ProductCategory, YEAR(SaleDate) AS SaleYear, SUM(Amount) AS TotalRevenue,
COUNT(OrderByID) AS TotalOrders, AVG(Amount) AS AvgOrderValue FROM SALES_FACT SF JOIN DIM_REGION
DR ON SF.RegionID = DR.RegionID JOIN DIM_PRODUCT DP ON SF.ProductID = DP.ProductID WHERE
YEAR(SaleDate) BETWEEN 2022 AND 2025 GROUP BY Region, ProductCategory, YEAR(SaleDate) ORDER BY
TotalRevenue DESC;
```

3.4 When to Use OLTP vs OLAP

Scenario	Use OLTP or OLAP?	Reason
Processing customer orders in an e-commerce app	OLTP	Real-time, short transactions
Generating quarterly sales report for management	OLAP	Historical aggregation
ATM cash withdrawal system	OLTP	Speed & ACID compliance
Predicting next year's product demand	OLAP	Trend analysis over large data
Hospital patient admission system	OLTP	Frequent updates, current data
Company-wide BI dashboard	OLAP	Multi-dimensional analysis

Modern systems often use BOTH: an OLTP database handles live operations, while an ETL (Extract-Transform-Load) pipeline feeds a separate OLAP data warehouse for analytics. This hybrid approach is the backbone of enterprise data architecture.

Summary

SQL Data Types	Define what kind of data each column stores. Choose wisely for performance and accuracy.
ER Diagrams	Visual tool to plan database structure before writing SQL. Maps entities, attributes, and relationships.
OLTP	Fast, transactional, normalized databases for daily operations. Uses ACID properties.
OLAP	Analytical, denormalized data warehouses for complex reporting and business intelligence.